

E-Commerce React Frontend Project

Requirements Gathering

Stakeholder Analysis · User Stories · Use Cases · System Requirements

1. Stakeholder Analysis

This section identifies all key stakeholders of the E-Commerce React Frontend application, their roles, primary needs, and level of influence on the project.

Stakeholder Register

ID	Stakeholder	Type	Role	Influence
SH-01	Guest / Visitor	External	Browses products without registering	High
SH-02	Registered Customer	External	Logs in, purchases, manages wishlist & orders	High
SH-03	UX / UI Designer	Internal	Designs mockups, style guide & accessibility	Medium
SH-04	Frontend Developer	Internal	Implements React components & state management	High
SH-05	Backend / API Team	Internal	Provides REST APIs consumed by the frontend	High
SH-06	Academic Supervisor	External	Evaluates documentation, design & code quality	High

Detailed Stakeholder Profiles

Guest / Visitor · Influence: High

Unauthenticated site visitor

- Browse product listing without login
- View full product details & images
- Add items to cart (session-based)
- Quick access to search & filters

Registered Customer · Influence: High

Authenticated end user

- Secure login
- Persistent cart & wishlist across sessions
- Smooth multi-step checkout process
- Order history & tracking

Product Manager · Influence: High

Business owner / catalog admin

- Clear product categorisation & filtering
- Promotional banners & discount visibility
- Product availability indicators
- Customer-facing KPIs (conversion, bounce)

UX / UI Designer · Influence: Medium

Interface & experience designer

- Consistent component library (React)
- Accessible color contrast & typography
- Responsive layouts for all screen sizes
- User-tested interaction patterns

Frontend Developer · Influence: High

React implementation lead

- Well-defined API contracts (REST/JSON)
- Reusable component architecture
- State management strategy (Context/Redux)
- Clear routing & page structure

2. User Stories

User stories describe the desired functionality from the perspective of the end user. Each story follows the standard format: As a [role], I want to [goal], so that [benefit].

Authentication — Login & Sign Up

US-01 Register New Account Sign Up Page

User Story As a guest, I want to create a new account with my name, email, and password, so that I can access personalized features like wishlists, order history, and a persistent cart.

Acceptance Criteria

- Sign-up form includes: full name, email, password, and confirm password fields
- Real-time validation shows inline errors (invalid email format)
- Password field includes show/hide toggle
- On successful registration, user is automatically logged in and redirected to the home page
- Duplicate email shows a clear error message with a link to the login page

US-02 Log In to Account Login Page

User Story As a registered customer, I want to log in with my email and password, so that I can access my account, saved cart, wishlist, and order history.

Acceptance Criteria

- Login form includes email and password fields
- Incorrect credentials show a clear error message without revealing which field is wrong
- Password field includes show/hide toggle
- Successful login redirects the user to the previously visited page or the home page

US-03 Log Out of Account Global (Header)

User Story As a registered customer, I want to log out of my account, so that my account remains secure on shared or public devices.

Acceptance Criteria

- Log out option accessible from the user menu in the navigation header
- Clicking log out clears the session and redirects to the home page
- Cart and wishlist state is cleared from the UI after logout
- No sensitive user data remains accessible after logout

Product Browsing

US-04 Browse Product Listing Product List Page

User Story As a guest or registered customer, I want to view a paginated list of all available products with images, names, and prices, so that I can quickly scan the catalog and find items of interest.

Acceptance Criteria

- Products displayed in a responsive grid
- Each card shows product image, name, price, and rating
- Pagination or infinite scroll loads additional products

US-05 Search for Products Product List Page

User Story As a guest or customer, I want to search the product catalog by keyword from any page, so that I can locate specific products without browsing the entire catalog.

Acceptance Criteria

- Search bar visible in the global navigation header
- Empty-state message shown when no results are found
- Search term highlighted in matching product names

Product Details

US-06 View Product Details Product Details Page

User Story As a customer, I want to view a dedicated page for a product showing full details, multiple images, and reviews, so that I can make an informed purchase decision before adding to cart.

Acceptance Criteria

- Page shows image gallery with thumbnail navigation
- Full description, specifications, and availability status displayed
- Breadcrumb navigation links back to the product list

Cart & Wishlist

US-7 Add Product to Cart Product Details / List Page

User Story As a guest or registered customer, I want to add a product to my shopping cart from the product details or listing page, so that I can collect items I intend to purchase before checking out.

Acceptance Criteria

- 'Add to Cart' button triggers visual confirmation (toast/badge update)
- Cart item count in header updates immediately
- Guest cart is session-based; registered user cart persists across sessions

US-8 Manage Cart Items Cart Page

User Story As a customer, I want to view, update quantities, and remove items in my shopping cart, so that I have full control over what I intend to purchase before checkout.

Acceptance Criteria

- Cart page lists all items with image, name, price, and quantity control
- Quantity stepper allows increase/decrease with minimum of 1
- Remove button deletes item immediately
- Order subtotal updates in real time when quantities change
- Empty-cart state shown with a CTA to continue shopping

US-9 Add Product to Wishlist Wishlist Page

User Story As a registered customer, I want to save products to a wishlist from the listing or details page, so that I can track items I like and purchase them later.

Acceptance Criteria

- Heart icon on each product card toggles wishlist status
- Unauthenticated users are redirected to login when clicking the wishlist icon
- Wishlist persists across sessions for logged-in users

US-10 View & Manage Wishlist Wishlist Page

User Story As a registered customer, I want to view all wishlisted products and move them to the cart or remove them, so that I can easily act on products I have saved for later.

Acceptance Criteria

- Wishlist page displays all saved products in a responsive grid
- 'Add to Cart' and 'Remove' actions available on each wishlist card
- Empty wishlist state shown with link to browse products

Checkout

US-11 Proceed to Checkout Checkout Page

User Story As a registered customer, I want to proceed from the cart to a multi-step checkout process, so that I can complete my purchase smoothly with clear guidance at each step.

Acceptance Criteria

- Checkout requires authentication; guests prompted to log in or register
- Three clear steps shown: Shipping → Payment → Review & Confirm
- User can navigate back to previous steps without losing entered data

US-12 Enter Shipping Information Checkout Page

User Story As a customer, I want to enter and save my shipping address during checkout, so that my order is delivered to the correct location without re-entering details next time.

Acceptance Criteria

- Address form includes: full name, street, city, postal code, country
- Saved addresses from profile auto-populate the form

US-12 Enter Shipping Information Checkout Page

- Real-time validation highlights errors before advancing to next step

User Stories Summary

ID	Title	Page	Stakeholder
US-01	Register New Account	Auth / Sign Up	Guest
US-02	Log In to Account	Auth / Login	Registered Customer
US-03	Log Out of Account	Auth (Global Header)	Registered Customer
US-04	Browse Product Listing	Product List	Guest / Customer
US-05	Search for Products	Product List	Guest / Customer
US-06	View Product Details	Product Details	Customer
US-7	Add Product to Cart	Product Details / List	Guest / Customer
US-8	Manage Cart Items	Cart	Customer
US-9	Add Product to Wishlist	Wishlist	Registered Customer
US-10	View & Manage Wishlist	Wishlist	Registered Customer
US-11	Proceed to Checkout	Checkout	Registered Customer
US-12	Enter Shipping Information	Checkout	Customer

3. Use Case Analysis

This section defines all use cases for the e-commerce system, describing interactions between actors and the system across all pages.

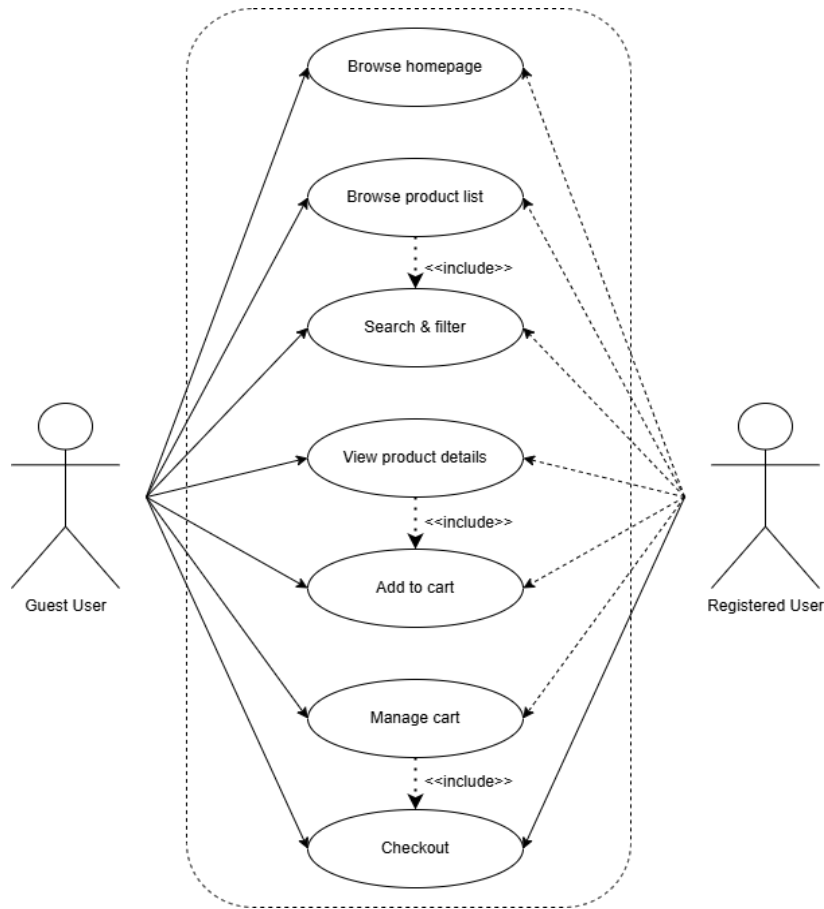
3.1 Actors

Two primary actors interact with the e-commerce system:

- Guest User: An unauthenticated visitor who can browse products, view details, and add to cart but cannot complete checkout.
- Registered User: An authenticated customer who has full access to all pages including Checkout.

3.2 Use Case Summary

UC ID	Use Case Name	Actor(s)	Page
UC-01	Browse Homepage	Guest / Registered User	Homepage
UC-02	Browse Product List	Guest / Registered User	Product List
UC-03	Search & Filter Products	Guest / Registered User	Product List
UC-04	View Product Details	Guest / Registered User	Product Details
UC-05	Add to Cart	Guest / Registered User	Product Details
UC-06	Manage Cart	Guest / Registered User	Cart
UC-07	Checkout & Place Order	Registered User	Checkout



3.3 Use Case Descriptions

UC-01 – Browse Homepage	
Actor	Guest / Registered User
Description	The user visits the homepage to explore the website, view promotions, and navigate to product sections.
Precondition	User has a working internet connection and opens the application URL.
Main Flow	1. System renders the homepage with a full-width hero banner. 2. System displays a featured products section with product cards. 3. System renders the navigation bar with links to product categories. 4. User clicks a product card or category link to navigate further.
Alt. Flow	1. Hero banner image fails to load: system shows a fallback-colored banner. 2. Featured products API fails: system shows a loader and error message.
Postcondition	User is directed to the Product List or Product Details page.

UC-02 – Browse Product List

Actor	Guest / Registered User
Description	The user views a paginated list of products within a selected category.
Precondition	User has navigated to the Product List page via menu or homepage link.
Main Flow	1. System fetches and displays product cards (image, name, price). 2. System renders a pagination control at the bottom. 3. User scrolls through the list and clicks a product card. 4. System navigates user to the Product Details page.
Alt. Flow	1. No products found in category: system displays 'No results found' message. 2. API timeout: system shows cached results or a retry button.
Postcondition	User views a product list or navigates to a product detail page.

UC-03 – Search & Filter Products

Actor	Guest / Registered User
Description	The user searches for products by keyword and applies filters and sort options to narrow results.
Precondition	User is on the Product List page.
Main Flow	1. User types a keyword into the search bar; system updates results in real-time. 2. User selects a category filter from the sidebar. 3. System re-renders the product grid with matching results.
Alt. Flow	1. No results match the search/filter combination: system displays 'No products found'.
Postcondition	Product list is updated to reflect the applied search and filter criteria.

UC-04 – View Product Details

Actor	Guest / Registered User
Description	The user views information about a specific product, including images, product description & price.
Precondition	User has clicked on a product card from the Product List or Homepage.
Main Flow	1. System loads the Product Details page for the selected product ID. 2. System renders an image gallery with thumbnail navigation. 3. System displays product name, price, availability & description. 4. User can select a quantity and click 'Add to Cart'.
Alt. Flow	1. Product is out of stock: 'Add to Cart' button is disabled; system shows 'Out of Stock' badge. 2. Image fails to load: system shows a message: 'image unavailable to view.'
Postcondition	User has reviewed the product and may proceed to add it to cart.

UC-05 – Add to Cart

Actor	Guest / Registered User
Description	The user adds a selected product with a specified quantity to their shopping cart.
Precondition	User is on the Product Details page and the product is in stock.
Main Flow	1. User selects a quantity. 2. User clicks the 'Add to Cart' button. 3. System updates the cart state. 4. Navbar cart icon badge updates to reflect new item count.
Alt. Flow	1. Requested quantity exceeds stock: system shows a warning: 'Only X items left in stock.' 2. Item already in cart: system updates quantity instead of duplicating the entry.
Postcondition	Item is added to cart. User may continue shopping or navigate to cart.

UC-06 – Manage Cart

Actor	Guest / Registered User
Description	The user reviews the contents of their cart, adjusts quantities, removes items, and views the running subtotal.
Precondition	User has at least one item in the cart and navigates to the Cart page.
Main Flow	1. System renders a list of cart items with image, name, unit price, and quantity. 2. User increments or decrements item quantity using '+' / '-' controls. 3. System recalculates and displays the updated subtotal in real time. 4. User clicks the remove icon to delete an item from the cart. 5. User clicks 'Proceed to Checkout' to continue.
Alt. Flow	1. Cart is empty: system displays 'Your cart is empty' with a link to the Product List. 2. Quantity updated to 0: system auto-removes the item from cart.
Postcondition	Cart reflects desired items and quantities. User proceeds to Checkout or continues shopping.

UC-07 – Checkout & Place Order

Actor	Registered User
Description	The registered user completes a multi-step checkout process: shipping address, payment details, order review, and order confirmation.
Precondition	User is authenticated and has at least one item in the cart.
Main Flow	1. System renders Step 1 (Shipping Info): user enters name, address, city.

UC-07 – Checkout & Place Order	
Alt. Flow	2. User clicks 'Continue'; system validates all fields. 3. System renders Step 2 (Payment): user enters card details. 4. User clicks 'Continue'; system validates payment fields. 5. System renders Step 3 (Order Review): itemized list, shipping cost & total. 6. User clicks 'Place Order'; system submits order to the backend API. 7. System renders the Confirmation page with order ID and estimated delivery date.
	1. Validation fails on any step: system highlights invalid fields with error messages. 2. Payment is declined: system shows payment error and returns to Step 2. 3. API submission fails: system shows a retry option and preserves entered data.
Postcondition	Order is placed successfully. Cart is cleared.

4. Functional Requirements

The table below defines the specific behaviors and features the e-commerce system must provide.

FR ID	Feature	Description
FR-01	Homepage Display	Render hero banner, featured products, promotional sections, and navigation bar
FR-02	Product Listing	Display paginated list of products with image, name, price, and rating
FR-03	Search	Allow keyword search with real-time results
FR-04	Product Details	Show full product info: images (gallery), description, specs & price
FR-05	Add to Cart	Add items to cart with quantity selection; update cart badge in navbar
FR-06	Cart Management	View cart items, update quantities, remove items, view subtotal
FR-07	Checkout Flow	Multi-step checkout: shipping info → payment → order review → confirmation
FR-8	Responsive Layout	All pages adapt across desktop, tablet, and mobile

5. Non-Functional Requirements

The table below specifies the quality attributes and operational constraints the system must satisfy.

NFR ID	Requirement	Description
NFR-01	Fast Loading	Pages should load quickly so users are not kept waiting when browsing products or checking out.
NFR-02	Mobile Responsive	The website must look and work well on all screen sizes, including phones and tablets.
NFR-03	Secure Login UI	The login and checkout forms must be secure, protecting user credentials and payment information.
NFR-04	Easy Navigation	Users should be able to find products, access the cart, and move between pages without confusion.
NFR-05	Simple & Clean Design	The interface should be visually clear and uncluttered so users can focus on shopping.
NFR-06	Always Available	The website should be up and running at all times with minimal downtime.
NFR-07	Error Messages	When something goes wrong, the system should show a clear warning message.